# LECTURE 9

## 6.5. Isomorphism of rings.

**Definition 6.21.** Let $R$ and $S$ be two rings. A bijective map $f : R \to S$ is called an *isomorphism* from $R$ to $S$ if

$$f(a \oplus_R b) = f(a) \oplus_S f(b)$$

and

$$f(a \odot_R b) = f(a) \odot_S f(b)$$

for all $a, b \in R$.

The rings $R$ and $S$ are then called *isomorphic rings*. We write $R \cong S$.

**Example 6.22.** The map $\sigma : \mathbb{C} \to \mathbb{C}$ defined by $\sigma(x+iy) = x-iy$ (i.e. complex conjugation) is a ring isomorphism of $\mathbb{C}$. To see this, we take complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, and consider

$$\begin{aligned}\sigma(z_1 + z_2) &= \sigma(x_1 + x_2 + i(y_1 + y_2)) \\ &= x_1 + x_2 - i(y_1 + y_2) \\ &= x_1 - iy_1 + x_2 - iy_2 \\ &= \sigma(z_1) + \sigma(z_2)\end{aligned}$$

and

$$\begin{aligned}\sigma(z_1 z_2) &= \sigma(x_1x_2 - y_1y_2 + i(x_1y_2 + y_1x_2)) \\ &= x_1x_2 - y_1y_2 - i(x_1y_2 + y_1x_2) \\ &= (x_1 - iy_1)(x_2 - iy_2) \\ &= \sigma(z_1)\sigma(z_2),\end{aligned}$$

hence $\sigma$ preserves both operations.

To show that $\sigma$ is a bijection, it suffices to note that $\sigma^2$ is the identity map on $\mathbb{C}$. Hence $\sigma = \sigma^{-1}$. Since the inverse map to $\sigma$ exists, it is clear that $\sigma$ is a bijection.

Therefore we have estabished that $\sigma$ is a ring isomorphism.

**Example 6.23.** Consider $M = \left\{ \begin{pmatrix} a & -b \\ 0 & a \end{pmatrix} \mid a, b, \in \mathbb{R} \right\}$ with ordinary matrix addition and multiplication. The set $M$ is a ring (check that for $A, B \in M$, we have $A + B \in M$ and $A \cdot B \in M$). Now define $f : M \to \mathbb{C}$ by

$$f\left( \begin{pmatrix} a & -b \\ 0 & a \end{pmatrix} \right) = a + ib.$$

It is clear that $f$ is surjective since given any $a + ib \in \mathbb{C}$ we get this element as the image of $\begin{pmatrix} a & -b \\ b & a \end{pmatrix} \in M$.

Also $f$ is injective because if $f(A) = 0$ then $a = b = 0$, and hence $A = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, the zero matrix of $M$.

This shows that $f$ is bijective.

Now we check that $f$ preserves the ring operations. For this, let $A = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ and $C = \begin{pmatrix} c & -d \\ d & c \end{pmatrix}$. We have

$$f(A+C) = f\left(\begin{pmatrix} a+c & -(b+d) \\ b+d & a+c \end{pmatrix}\right) = (a+c) + i(b+d)$$

$$f(A)+f(C) = f\left(\begin{pmatrix} a & -b \\ b & a \end{pmatrix}\right) + f\left(\begin{pmatrix} c & -d \\ d & c \end{pmatrix}\right) = (a+ib)+(c+id) = (a+c)+i(b+d)$$

so we have $f(A+C) = f(A)+f(C)$.

Likewise

$$\begin{aligned} f(A\cdot C) &= f\left(\begin{pmatrix} a & -b \\ b & a \end{pmatrix}\cdot\begin{pmatrix} c & -d \\ d & c \end{pmatrix}\right) = f\left(\begin{pmatrix} ac-bd & -(ad+bc) \\ ad+bc & ac-bd \end{pmatrix}\right) \\ &= (ac-bd)+i(ad+bc) \end{aligned}$$

$$f(A)\cdot f(C) = f\left(\begin{pmatrix} a & -b \\ 0 & a \end{pmatrix}\right)\cdot f\left(\begin{pmatrix} c & -d \\ 0 & c \end{pmatrix}\right) = (a+ib)\cdot(c+id) = (ac-bd)+i(ad+bc).$$

We have now seen that $f(A\cdot C) = f(A)\cdot f(C)$. We conclude from all the above that $M$ and $\mathbb{C}$ are isomorphic.

**Example 6.24** (Non-example)**.** Let $\theta$ be the map from the ring $\mathbb{Z}$ of integers to the ring $2\mathbb{Z}$ of even integers defined by $\theta(n) = 2n$ for each $n \in \mathbb{Z}$. Addition is preserved from $\mathbb{Z}$ to $2\mathbb{Z}$ because

$$\theta(m+n) = 2(m+n) = 2m+2n = \theta(m)+\theta(n)$$

for all $m, n \in \mathbb{Z}$.

It is injective, because if $\theta(m) = \theta(n)$ for $m, n \in \mathbb{Z}$, then $2m = 2n$ and so $m = n$.

It is surjective, because for any $k \in 2\mathbb{Z}$, we can write $k = 2m$ for some $m \in \mathbb{Z}$. Therefore $\theta(m) = 2m = k$ and so $m$ is a pre-image of $k$ under $\theta$.

But $\theta$ is not a ring isomorphism because it does not preserve multiplication:

$$\theta(mn) = 2mn$$

but

$$\theta(m)\theta(n) = (2m)(2n) = 4mn$$

for $m, n \in \mathbb{Z}$.

### 6.6. **Fermat's Little Theorem.**

**Theorem 6.25.** *Let $F$ be a finite field with $q$ elements. If $a \in F$ with $a \neq 0$, then $a^{q-1} = 1$ in $F$.*

*Proof.* Look at $F^* = \{x \in F \mid x \neq 0\} = \{x_1, x_2, \ldots, x_{q-1}\}$. Let $g : F^* \to F^*$ be given by $x_i \mapsto ax_i$. This is a bijective map (exercise). Hence $ax_1$ , $ax_2$, …, $ax_{q-1}$ is just the list $x_1$, $x_2$, …, $x_{q-1}$ in a different order. Therefore

$$(ax_1)(ax_2)\cdots(ax_{q-1}) = x_1x_2\cdots x_{q-1},$$

which is equivalent to

$$a^{q-1}x_1x_2\cdots x_{q-1} = x_1x_2\cdots x_{q-1}.$$

As $x_1x_2\cdots x_{q-1}$ is invertible, it follows that $a^{q-1}=1$, as required. □

In the special case where $F=\mathbb{Z}_p$ for $p$ a prime, we get $a^{p-1}\equiv 1 \pmod p$ for any $a$ not divisible by $p$. This is called Fermat's Little Theorem:

**Corollary 6.26** (Fermat's Little Theorem)**.** *For $a\in\mathbb{Z}$ not divisible by $p$, we have $a^{p-1}\equiv 1 \pmod p$.*

**Example 6.27.** Compute $19^{34} \pmod{11}$.

Solution: By Fermat's Little Theorem, we have $19^{10}\equiv 1 \pmod{11}$. Now

$$19^{34} = 19^{3\cdot 10+4} = (19^{10})^3\cdot 19^4 \equiv 1^3\cdot 19^4 \equiv 8^4 \equiv (64)^2 \equiv (-2)^2 \equiv 4,$$

where here we have used the fact that $19\equiv 8 \pmod{11}$. So the answer is $19^{34}\equiv 4 \pmod{11}$.

An alternative way to phrase Fermat's Little Theorem is $a^{m(p-1)+1}\equiv a$ for all integers $a$ and positive integers $m$. Using this, one can generalise Fermat's Little Theorem to product of primes:

**Theorem 6.28.** *Let $p$ and $q$ be different primes, let $a$ be an integer and $m$ a positive integer. Then*

$$a^{m(p-1)(q-1)+1}\equiv a \pmod{pq}.$$

*Proof.* For convenience, write $b=a^{m(p-1)(q-1)+1}-a$. We observe that it suffices to show that $p\mid b$ and $q\mid b$, as then it follows that $pq\mid b$, and so we are done.

Working modulo $p$, we have that for $p\nmid a$ that $a^{p-1}\equiv 1 \pmod p$ by Fermat's Little Theorem, and hence

$$a^{m(p-1)(q-1)+1} = \left(a^{p-1}\right)^{m(q-1)}\cdot a \equiv 1^{m(q-1)}\cdot a = a,$$

and so

$$a^{m(p-1)(q-1)+1}-a\equiv 0 \pmod p \quad\iff\quad p\mid (a^{m(p-1)(q-1)+1}-a).$$

If $p\mid a$, then clearly $p\mid (a^{m(p-1)(q-1)+1}-a)$.

An analogous argument shows that $q\mid (a^{m(p-1)(q-1)+1}-a)$, and hence the result follows. □

**Example 6.29** (RSA-system in cryptography)**.** RSA stands for Rivest, Shamir and Adleman, and this system was established in 1978.

If for example the bank wants many customers to send secret information they can use the following idea:

Start out with two large primes $p,q$. Let $n=p\cdot q$. Choose some $d$ with $(d,(p-1)(q-1))=1$ and compute the inverse $e$ of $d$ in $\mathbb{Z}_{(p-1)(q-1)}$. Now $n$ and $e$ are made public but $p$, $q$ and $d$ are secret.

To encode a number $C$, for $1\le C\le n$, compute $D=C^e \pmod n$ and send $D$ to the bank. The bank knows $d$ and can then compute

$$D^d = (C^e)^d = C^{ed} = C^{1+k(p-1)(q-1)} \equiv C \pmod n,$$

where the last equality use the fact that $e$ and $d$ are inverses in $\mathbb{Z}_{(p-1)(q-1)}$, and the equivalence uses Theorem 6.28. The message is then decoded.

If you can find the factors $p$ and $q$ of $n$, you will know $(p-1)(q-1)$ and can compute $d$. Hence you can break the code. But factorisation is very computationally demanding, so with large enough $p$, $q$ the bank should be safe.

6.7. **Chinese remainder theorem.** Consider the following example of solving systems of congruences.

**Example 6.30.** Solve the following system of congruences:

$$\begin{aligned} x &\equiv 3 \pmod 5 \\ x &\equiv 1 \pmod 7 \\ x &\equiv 2 \pmod{11} \end{aligned}$$

<u>Solution:</u> A pair of congruences can be turned into a Diophantine equation. For example, a solution to

$$\begin{aligned} x &\equiv 3 \pmod 5 \\ x &\equiv 1 \pmod 7 \end{aligned}$$

is equivalent to the existence of $a, b \in \mathbb{Z}$ such that $x = 3 + 5a = 1 + 7b$. We can find all $a, b$ with

$$3 + 5a = 1 + 7b \Longleftrightarrow 7b - 5a \overset{(*)}{=} 2.$$

To solve $(*)$, first note that $(5, 7) = 1$. Then using Euclid's algorithm:

$$\begin{aligned} 7 &= 1 \cdot 5 + 2 \\ 5 &= 2 \cdot 2 + 1 \end{aligned}$$

and backtracking yields

$$\begin{aligned} 1 &= 5 - 2 \cdot 2 \\ &= 5 - 2(7 - 1 \cdot 5) \\ &= 3 \cdot 5 - 2 \cdot 7. \end{aligned}$$

Then

$$\begin{aligned} 2 \times \big(1 &= 3 \cdot 5 - 2 \cdot 7\big) \\ 2 &= 6 \cdot 5 - 4 \cdot 7 \end{aligned}$$

gives a solution

$$a = -6, \quad b = -4.$$

Then all solutions are given by

$$a = -6 + 7k, \quad b = -4 + 5k \quad \text{for } k \in \mathbb{Z}.$$

For $x$ this results in

$$\begin{aligned} x &= 3 + 5a \\ &= 3 + 5(-6 + 7k) \\ &= -27 + 35k \end{aligned}$$

(or $x = 1 + 7b = 1 + 7(-4 + 5k) = -27 + 35k$). The solution is therefore $x \equiv -27 \pmod{35} \equiv 8 \pmod{35}$.

But the above method only deals with two congruences. For three or more congruences, we must repeat the process above or use the next result.

**Theorem 6.31** (Chinese Remainder Theorem). *Let $n_1, n_2, \ldots, n_k \in \mathbb{Z}$ be pairwise relatively prime. Then the system of congruences*

$$\begin{aligned} x &\equiv a_1 \pmod{n_1} \\ x &\equiv a_2 \pmod{n_2} \\ &\vdots \\ x &\equiv a_k \pmod{n_k} \end{aligned}$$

*has a unique solution modulo $n = n_1 n_2 \cdots n_k$.*

*Proof.* First we find a solution $x$. Let $N_i = \frac{n}{n_i}$. Then $(N_i, n_i) = 1$ so using Euclid's algorithm we can find $s_i$ and $t_i$ with $1 = s_i N_i + t_i n_i$. Multiplying by $a_i$ gives

$$a_i = a_i s_i N_i + a_i t_i n_i.$$

Then we can see that

$$a_i s_i N_i \equiv \begin{cases} a_i \pmod{n_i}, \\ 0 \pmod{n_j} \text{ for } j \neq i. \end{cases}$$

Let

$$x = a_1 s_1 N_1 + a_2 s_2 N_2 + \cdots + a_k s_k N_k.$$

Regarding $x \pmod{n_i}$ we only get

$$a_i s_i N_i = a_i - a_i t_i n_i \equiv a_i.$$

So $x$ is a solution. It is also easy to see that $x + kn$, for $k \in \mathbb{Z}$, are solutions too.

*Uniqueness of the solution:* Assume that $x$ and $x'$ are both solutions to the system of congruences. Then modulo $n_i$ we have $x - x' \equiv a_i - a_i = 0$, which implies that $n_i \mid (x - x')$. This holds for $i \in \{1, 2, \ldots, k\}$. Hence

$$(n = n_1 n_2 \cdots n_k) \mid (x - x').$$

In other words, we have $x \equiv x' \pmod{n}$. [Note that in this last step, we are using the fact that the $n_i$ are pairwise coprime.] □

The proof also provides a method for finding the solution, as seen below.

**Example 6.32.** Solve:

$$\begin{aligned} x &\equiv 3 \pmod{5} \\ x &\equiv 1 \pmod{7} \\ x &\equiv 2 \pmod{11} \end{aligned}$$

Solution: Here $n = 5 \cdot 7 \cdot 11 = 385$. We will find a unique solution modulo 385.

$$\begin{aligned} n_1 = 5, \quad N_1 &= \frac{5 \cdot 7 \cdot 11}{5} = 7 \cdot 11 = 77, \\ n_2 = 7, \quad N_2 &= \frac{5 \cdot 7 \cdot 11}{7} = 5 \cdot 11 = 55, \\ n_3 = 11, \quad N_3 &= \frac{5 \cdot 7 \cdot 11}{11} = 5 \cdot 7 = 35. \end{aligned}$$

To compute $a_1 s_1 N_1$, note that $(5, 77) = 1$, so by Euclid's algorithm,

$$1 = 5 - 2 \cdot 2 = 5 - 2(77 - 15 \cdot 5) = \underbrace{31 \cdot 5}_{t_1 n_1} \underbrace{-2 \cdot 77}_{s_1 N_1},$$

and so $a_1 s_1 N_1 = 3(-2)(77) = -462$.

To compute $a_2 s_2 N_2$, as $(7, 55) = 1$, Euclid's algorithm gives $55 = 7 \cdot 8 - 1$, and so

$$1 = 8 \cdot 7 - 55$$

and so $t_2 = 8$ and $a_2 = 1$, $s_2 N_s = -55$, which gives $a_2 s_2 N_2 = 1(-1)(55) = -55$.

Lastly we compute $a_3 s_3 N_3$. Also $(11, 35) = 1$, and we have

$$35 = 3 \cdot 11 + 2$$
$$11 = 5 \cdot 2 + 1$$

which gives

$$1 = 11 - 5 \cdot 2 = 11 - 5(35 - 3 \cdot 11) = \underbrace{16 \cdot 11}_{t_3 n_3} \underbrace{-5 \cdot 35}_{s_3 N_3}.$$

Thus $a_3 s_3 N_3 = 2(-5)(35) = -350$.

Putting this together we obtain

$$\begin{aligned} x &= a_1 s_1 N_1 + a_2 s_2 N_2 + a_3 s_3 N_3 \\ &= -462 - 55 - 350 \\ &= -77 - 55 + 35 \\ &= -97. \end{aligned}$$

[Check that $-97 \equiv 3 \pmod 5$, $-97 \equiv -20 \equiv 1 \pmod 7$ and $-97 \equiv 2 \pmod{11}$.] Hence the solutions of the system are those $x \in \mathbb{Z}$ satisfying $x \equiv -97 \pmod{385}$.